Can Satellites Replace the Field Plot? A Blind, Time-Matched Benchmark of Satellite-Only Forest Carbon Estimation Against Standard-Formula Ground Truth

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***Abstract*—Satellite platforms promise to replace field inventories in forest carbon crediting, but their estimates are rarely tested blind against the standard-formula answer, and two practical problems, clouds and confidence, are rarely quantified. We separate the question into two isolated grids. The right grid computes field ground truth step by step to SNI 7724, IPCC, CDM, and Verra rules, with a Monte Carlo uncertainty budget, and is sealed; the left grid sees only plot coordinates and dates, reads time-matched Sentinel-1/2, Landsat, PALSAR, and terrain data, and estimates carbon blind. A hash-chained commit-reveal ledger makes any change to predictions after the reveal detectable. On 476 stock plots, blind error was 45–49% relative RMSE: about 30% on 1 ha plots, over 100% on small Indonesian plots, with saturation near 300 $\mathrm{Mg\,ha^{-1}}$. A plot-level climatology of 22,697 Sentinel-2 acquisitions shows that only 21.5% see an Indonesian plot clear, yet a common scene-level filter discards 65% of those; a radar-led model lost two points without optical data. Conformalized quantile regression calibrated on synthetically clouded copies of its calibration plots (missing-data augmentation) kept 90% coverage under full cloud; a metadata confidence score carried no information about error beyond dataset differences. Project intervals that ignore correlated map error covered 74% of projects; as auxiliary data for field plots, the satellite was 1.4–2.3 times as efficient as field plots alone on large-plot datasets. For annual growth on 196 Bornean plots, no satellite estimate beat assuming no change. Satellites can map stock and save field plots; growth still needs remeasurement, and confidence must be calibrated, not graded.**

***Index Terms*—forest carbon, MRV, SNI 7724, VM0047, remote sensing, cloud cover, Sentinel, ALOS PALSAR, machine learning, spatial cross-validation, conformal prediction, uncertainty, model-assisted estimation, commit-reveal ledger, distributed processing**

I. Introduction

A. Two Ways to Measure Forest Carbon

Traditional MRV (measurement, reporting, and verification) is field based. Crews lay out plots, measure each stem's diameter D and height H, identify species to assign wood density ρ, and pass these inputs through a published allometric equation [1]. Carbon follows from a fixed carbon fraction and the CO_2 to C mass ratio. Methodologies such as Verra VM0047 prescribe this chain [2]; its output is deterministic and is the number on which credits are issued. Nobody can change the formula: it is the international standard. What is costly is the visit, especially in remote Indonesian peat swamps and hill forests.

System-based MRV replaces or supplements the visit with satellites and software. Optical sensors observe canopy greenness, radar observes canopy structure and moisture, and spaceborne lidar measures height [13]. Global biomass maps exist [15], [16], and commercial platforms, including our own NusaCarbon prototype, turn polygons into carbon numbers. The open question is how close such numbers are to what the field formula would have produced at the same place and time.

B. Why the Comparison Is Hard to Do Honestly

Three traps inflate reported accuracy. First, *leakage*: if a model is trained or tuned on the plots it is scored on, or on plots next to them, the score measures memory rather than skill; spatially blocked validation of large-scale biomass maps reveals much lower skill than random validation [23], [24]. Second, *temporal mismatch*: a plot measured in 2009 and an image from 2023 describe different forests. Third, *scale*: a

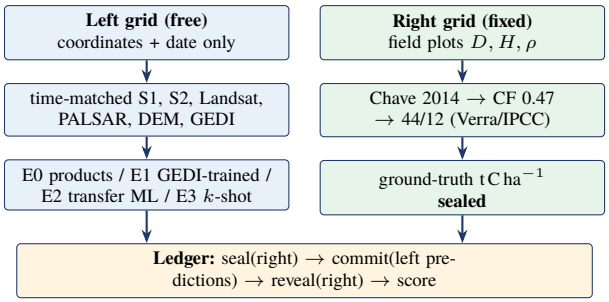


Fig. 1. Blind two-grid design. Only the public manifest (plot id, coordinates, date, radius) crosses from right to left.

plot is a fraction of a hectare while satellite pixels range from 10 m to 4.6 km. A fair test must isolate the estimator from the answer, match time explicitly, and report how error changes with scale.

Our own platform illustrates the need. An audit of the NusaCarbon code base found that its forest and mangrove figures depended mostly on default constants (for example, a fixed sequestration rate and tree density bins from NDVI) rather than on satellite measurement, that clouds were screened only at scene level, that the reported confidence was a grade computed from image metadata, and that no estimate had been compared with field data. This paper builds the comparison that was missing, and tests the cloud handling and the confidence statement as carefully as the estimate itself.

C. Two Isolated Grids

We separate the problem into two grids (Fig. 1). The **right grid** is fixed: published field plots, the standard formula, a carbon density in t C ha^{-1}. We never modify it. The **left grid** is free: any satellite data, any algorithm, any machine-learning method, subject to one rule: it never sees the right grid's outcomes for the plots it is scored on. A ledger enforces the rule and makes it auditable.

D. Research Questions

RQ1 (blind accuracy; research methodology). How far does a blind, satellite-only estimate of aboveground carbon stock deviate from standard-formula field ground truth, and how does the deviation depend on the temporal gap between field and satellite observations and on the area over which it is aggregated?

RQ2 (closing the gap; AI). Which learning strategy brings the satellite estimate closest to the field answer: off-the-shelf global products, a model trained only on spaceborne lidar, a model transferred from other regions, or a model calibrated with k local field plots? Are its prediction intervals honest?

RQ3 (clouds and confidence; AI and methodology). How often do clouds hide an Indonesian plot from optical satellites, how much accuracy does that cost, and can the estimate and its prediction interval stay honest when they do? At the scale of a project, does the stated uncertainty protect against over-crediting, and how many field plots does a satellite save?

RQ4 (sequestration). Can the annual change in carbon stock, the quantity credits are issued on, be estimated blind from satellites, and how certain is the field answer itself?

RQ5 (distributed systems). Can time-matched extraction for hundreds of plots and sensors be sharded, balanced, and made fault tolerant with bit-identical outputs, and can a hash-chained ledger make the order of prediction and reveal verifiable?

E. Contributions

(i) A blind, time-matched evaluation protocol with a commit-reveal ledger, reusable for any MRV estimator. (ii) A right grid built step by step to SNI 7724, IPCC, CDM, Verra, and Gold Standard rules, with a Monte Carlo uncertainty budget, on 476 stock plots (148 Indonesian plots recomputed tree by tree) and 223 day-dated repeat-census plots in Borneo. (iii) An estimator ladder from zero field data to k local plots, with spatially blocked validation. (iv) A plot-level Sentinel-2 clear-sky climatology for Indonesian forest plots and a measured error rate of the scene-level cloud filter. (v) A test of missing-data-augmented conformal intervals [72] under cloud-driven, non-random loss of the optical block with out-of-region calibration. (vi) An empirical coverage test of project-level intervals under spatial transfer, and a simulation of how often the VM0047 uncertainty deduction still over-credits. (vii) A sharded, idempotent extraction runner with measured scaling and fault tolerance.

What is not new. We propose no new estimator or interval method. Conformalized quantile regression [60], its missing-data augmentation [72], modality dropout [58], model-assisted estimation [71], [73], and the effects of plot size, saturation, allometric choice, and root-to-shoot choice [47], [67], [74], [75] are established. Hash chains, sealed test sets, and commit-reveal are decades old [27], [28], [76]. The contribution is a blind, standards-anchored evaluation that tests these tools together on Indonesian and Bornean plots, under the failure modes an MRV platform meets: clouds, small plots, spatial transfer, and the deduction rules of the registries.

II. Related Work

A. Allometry and Standards

Pantropical allometry links D, H, and ρ to aboveground biomass (AGB) [1]; the BIOMASS package propagates its uncertainty [11]. Verra VM0047 and the HCSA toolkit specify sampling, measurement thresholds, and plot designs [2], [3]. Investigations of REDD+ crediting show that unverifiable baselines produce over-crediting [12], which motivates auditable estimates.

B. Remote Sensing of Biomass

Optical reflectance saturates over dense canopies. C-band radar mostly sees the upper canopy; L-band radar penetrates further and tracks woody biomass up to moderate densities [18], but its backscatter can *decrease* in very dense forest [19]. Spaceborne lidar from GEDI samples canopy height in 25 m footprints and is converted to biomass by

footprint models calibrated on field data [13], [14]. Global AGB maps combine these signals: ESA CCI Biomass [15], the 2010 harmonized map of Spawn *et al.* [16], and the annual Chloris product [17]. Canopy height alone predicts plot carbon well where lidar is available [20].

C. Machine Learning and Honest Validation

Gradient-boosted trees are strong tabular regressors [21]; models fitted on log biomass need a retransformation correction [22]. Spatial autocorrelation makes random cross-validation optimistic; blocked or leave-region-out designs are required [23], [24]. Conformal prediction yields distribution-free intervals with finite-sample coverage [25]. Agreement is assessed with RMSE, bias, and Lin's concordance [26].

D. Clouds and Uncertainty

Southeast Asia is cloudy on at least 80% of days [48], and scene metadata suggest at most one cloud-free Sentinel-2 scene per year over much of tropical Indonesia [49]. The combined Sentinel-2 and Landsat revisit is 2.9 days [50]. Cloud masks such as Fmask and Cloud Score+ reach about 90% accuracy, with thin cloud and shadow the main failures [51]–[53]. Flores-Anderson *et al.* map the frequency of clear 30 m observations across the tropics [77]. Workarounds include best-available-pixel compositing [54], harmonized Landsat–Sentinel data [55], SAR–optical fusion [56], and deep SAR-to-optical cloud removal [57]. On the uncertainty side, split conformal prediction [59], conformalized quantile regression [60], normalized [61] and Mondrian [62] variants, and weighted or spatial conformal prediction under shift [63], [64] give distribution-free intervals; Zaffran *et al.* extend them to missing covariates by augmenting the calibration set with the test point's missingness pattern [72]. Conformal intervals have been applied to forest-attribute maps from Sentinel-2 [78] and to Earth-observation models in general [79]. For biomass maps, the CEOS protocol [65] and Duncanson *et al.* [66] recommend plots of at least 1 ha and explicit error propagation, and Araza *et al.* separate map error from plot error [67]. Model-assisted and model-based estimators that combine remote sensing with field samples are standard in forest inventory [70], [71], [73]; their efficiency grows with plot size [80], and model-based variance estimators under-cover when residual spatial correlation or model bias is ignored [81], [82].

E. Integrity and Distributed Processing

Hash-chained time-stamping makes a document's existence at a time verifiable [27]; commit-reveal protocols bind a party to a value before another party reveals theirs [28]. Large geospatial workloads are processed as independent map tasks [29]; consistent hashing places keys on workers with minimal remapping [30], and Graham's longest-processing-time rule bounds makespan within $4/3$ of optimal [31]. Cloud catalogues such as the Microsoft Planetary Computer [32] and Google Earth Engine expose these archives through APIs.

TABLE I
RIGHT-GRID FIELD DATASETS (STOCK)

Source	Region	Plots	Blocks	Date basis
HCSA [4]	Indonesia (6 prov.)	148	6	release 2023
GEO-TREES [5]	Pantropical	142	15	census year
Labrière [6]	Fr. Guiana, Gabon	186	6	campaign year
Total		**476**	**27**	

III. METHODOLOGY

A. Right Grid: Field Ground Truth by the Standards

The right grid is the number a carbon standard would accept. We therefore build it the way the standards prescribe, from sampling design to deduction, and keep the deterministic point value separate from its uncertainty. Table I lists the plot sources and Table II the frameworks we follow.

Step 1, tree. Aboveground biomass of a tree with diameter D (cm), height H (m), and wood density ρ (g cm^{-3}) follows Chave 2014 Eq. 4 [1],

$$\mathrm{AGB}_i = 0.0673\,(\rho_i D_i^2 H_i)^{0.976}\ [\mathrm{kg}], \qquad \sigma_{\ln} = 0.357, \quad (1)$$

where $\sigma_{\ln}$ is the residual standard error on the log scale. Where height is not measured, a local model $\ln H = a + b \ln D$ with Duan correction replaces it (SAFE: $a = 0.549$, $b = 0.656$, residual SD 0.348); Chave Eq. 7 with the environmental stress term E is the fallback where no local heights exist. Wood density is the mean of the Global Wood Density Database [9], [10] at the finest level available, species, then genus, then family, preferring South-East Asian records. As a check on model choice we also apply the East-Kalimantan equation of Basuki *et al.* [43], $\ln \mathrm{AGB} = -0.744 + 2.188 \ln D + 0.832 \ln \rho$ (correction factor 1.047). Palms are excluded: Chave's equations are for dicotyledonous trees, and palm allometry depends on stem height, not diameter [44], [45].

Step 2, plot. Tree biomass is expanded by the area in which a stem of its size is counted. For the HCSA nested design [3],

$$\mathrm{AGB}_j = \sum_{i \in j} \frac{\mathrm{AGB}_i}{1000\, a(D_i)}, \quad a(D) = \begin{cases} 0.01, & 5 \le D < 15 \\ 0.05, & D \ge 15 \end{cases} \quad (2)$$

in Mg ha^{-1}; SNI 7724 uses the same form, $C_n = C_x/1000 \times 10{,}000/l_{\mathrm{plot}}$, with its own subplot areas [36].

Step 3, pools and carbon. The plot carbon density is

$$\begin{aligned} C_j &= \mathrm{CF}\,(1+R)\,\mathrm{AGB}_j + C_{\mathrm{DW},j} + C_{\mathrm{L},j} + C_{\mathrm{SOC},j}, \\ \mathrm{CO_2e}_j &= \tfrac{44}{12}\, C_j, \end{aligned} \quad (3)$$

with CF $= 0.47$ (IPCC 2006 Table 4.3, unchanged in 2019; SNI 7724; Verra) or 0.475 (Gold Standard). For the root-to-shoot ratio R the frameworks disagree: SNI 7724 Lampiran A adopts the IPCC 2006 tropical-rainforest value $R = 0.37$, while the 2019 Refinement gives 0.207 (SD 0.072) below 125 Mg ha^{-1} and 0.212 (SD 0.077) above for natural Asian tropical rainforest [33], [34]. Dead wood, litter, and soil carbon are not in the public records; omitting a pool can only

TABLE II
FIELD-INVENTORY RULES OF THE FRAMEWORKS APPLIED TO THE RIGHT GRID

Framework	Plot design (minimum stem size)	Confidence	Precision	Uncertainty rule
SNI 7724:2011 [36]	nested 2×2 (seedlings, litter), 5×5 (saplings 2–10 cm), 10×10 (poles 10–20 cm), 20×20 m (trees ≥20 cm); five pools, Tier 3	not stated	sampling error ≤20%	stratified systematic or random sampling
Indonesian NFI [37]	20 km grid; 100×100 m permanent plot of 16 units of 25×25 m; subplot radii 1/2/5 m; trees ≥20 cm	–	–	national FRL, Manuri allometry, CF 0.47
HCSA Toolkit v2.0 [3]	circular, $r = 5.64$ m for 5–14.9 cm, $r = 12.61$ m (0.05 ha) for ≥15 cm	90%	±10%	classes HDF, MDF, LDF, YRF, S, OL
IPCC 2006/2019 [33]–[35]	five pools; CF 0.47; root-to-shoot R by zone	95%	–	Approach 1 propagation or Approach 2 Monte Carlo
CDM AR-TOOL14/03 [38], [39]	stratified random plots	90%	±10%	stepped discount above 10%; sampling error only
Verra VM0047 v1.1 [2]	plot sampling, remeasured ≤5 yr	90%	–	deduct $U - 10\%$; zero credit if $U > 100\%$
Gold Standard A/R [40]	Winrock sourcebook plots; CF 0.475	90%	±20%	deduct the excess over 20%

lower a stock claim, which is the conservative direction. All comparisons with the left grid use AGB, which removes the R and CF choices from the benchmark; we report their effect separately.

Step 4, stratum and project. Following AR-TOOL14 Eqs. 14–17 [38], plots are grouped into strata h (the HCS classes) with weights w_h, and the project mean and its relative uncertainty at 90% are

$$\bar{b} = \sum_h w_h \bar{b}_h, \qquad u = \frac{t_{0.95,\,n-M}\sqrt{\sum_h w_h^2 s_h^2 / n_h}}{\bar{b}}, \tag{4}$$

with s_h^2 the within-stratum variance, n_h plots in stratum h, n plots in total, and M strata. The mapped stratum areas are not public, so the plot shares stand in for w_h. The number of plots needed for a margin E (AR-TOOL03 Eq. 2, Neyman allocation, iterated on t when $n < 30$) [39] is

$$n = \left(\frac{t_{0.95}}{E}\right)^2 \left(\sum_h w_h s_h\right)^2, \qquad n_h = n\,\frac{w_h s_h}{\sum_h w_h s_h}. \tag{5}$$

Step 5, uncertainty. AR-TOOL14 counts sampling error only. The IPCC asks for all sources, combined by Approach 1, $U = \sqrt{\sum_k U_k^2}$ for products and $U = \sqrt{\sum_k (U_k x_k)^2}/|\sum_k x_k|$ for sums, or by Monte Carlo (Approach 2) [35]. We use Approach 2 with 2,000 draws per tree and the error models of the BIOMASS package [11]: diameter error with SD $0.0062D + 0.0904$ cm for 95% of stems and 4.64 cm for a random 5% [42]; height error of 10% [42]; wood-density SD of 0.0708, 0.0941, and 0.1234 g cm^{-3} at species, genus, and family level; the allometric residual $\varepsilon_i \sim \mathcal{N}(0, \sigma_{\ln}^2)$; CF $\sim \mathcal{N}(0.4713, 0.0206^2)$ [46]; and R with its 2019 SD. The project error budget splits the variance of $\bar{b}$ into a sampling part (Eq. 4) and a measurement-plus-allometry part (the Monte Carlo variance of the same weighted mean).

Step 6, change and deduction. Sequestration uses the IPCC stock-difference method, $\Delta C = (C_{t_2} - C_{t_1})/(t_2 - t_1)$ (Vol. 4 Eq. 2.5), on the same plots at both dates [33]. Its uncertainty follows VM0047 v1.1 Eq. 28 [2],

$$U = \frac{T\sqrt{SE_0^2 + SE_1^2 - 2\rho\,SE_0 SE_1}}{\Delta C}, \qquad \mathrm{UNC} = \min\big(1, \max(0,\ U - 0.10)\big), \tag{6}$$

where SE_0 and SE_1 are the standard errors of the mean stock at the two dates, ρ is their correlation on permanent plots, and T is the two-tailed t value at $\alpha = 0.1$. Credits are multiplied by $1 - \mathrm{UNC}$. For stocks, we report the deductions of VM0047 ($u - 10\%$), AR-TOOL14 (a stepped share of u: 0 up to 10%, then 25, 50, 75, 100% of u at 15, 20, 30%), Gold Standard ($u - 20\%$), and VMD0017 (95% uncertainty minus 15%) [38], [40], [41]. In the Monte Carlo for change, a tree keeps the same allometric and height residual at both censuses, because the same stem is measured with the same equation; we also run the case in which they are redrawn.

Data screening. For HCSA, Step 1 is recomputed for every tree. Sixty-two records stated an area that contradicts Eq. 2 and were expanded by the protocol area. Whole plots were excluded, never imputed, if any record had a missing input, $H < 1.3$ m, conflicting coordinates, or $D > 200$ cm. The last rule caught plot RU09 (Riau), where a *Gluta renghas* was recorded at $D = 480$ cm, which matches a circumference typed into the diameter column ($480/\pi \approx 153$ cm); left in place it would have produced 4,426 Mg ha^{-1}. Of 185 plots, 148 were retained; the 15 records labelled as oil palm all fell in excluded plots. GEO-TREES publishes plot AGB from forest networks [5]; 154 plots with withheld AGB were dropped. Labrière *et al.* publish 1 ha units with AGB from Chave Eq. 4 and local height-diameter models, plus 95% credible intervals [6]; we use the 1 ha units only.

The outcome table is written to a sealed file. Only a manifest of plot id, region block, coordinates, date, date precision, and extraction radius is released to the left grid.

B. Temporal Matching

Field inventories record a date or, more often, a year; no public inventory records the minute of measurement, and optical satellites revisit every 5–16 days with frequent cloud. We therefore match by window: for a plot measured at t_j, the left grid uses only clear observations in $[t_j - W,\, t_j + W]$, records the offset in days of the closest one, and takes the median of up to three clear scenes. The default is $W = 90$ days; RQ1 varies $W \in \{30, 90, 180, 365\}$ on the Indonesian plots. Each record also carries its timestamp precision (*day*, *year*, or *campaign year*) so that the analysis can separate sensor noise from date uncertainty.

C. Left Grid: Satellite Features

For each plot the left grid reads, inside a square window around the plot centre: Sentinel-2 L2A reflectance in ten bands with scene-classification cloud masking and seven indices (NDVI, EVI, NDRE, NDMI, NBR, IRECI, SAVI); Landsat 5–9 Collection 2 surface reflectance with QA cloud masking; Sentinel-1 RTC backscatter (VV, VH, ratio); the ALOS PALSAR annual L-band mosaic (HH, HV); and Copernicus 30 m elevation and slope. Two global products are read as zero-shot baselines: Chloris annual AGB (2003–2019) [17] and the 2010 harmonized map [16]. GEDI L4A footprint biomass and L2A relative height [14] are read through Earth Engine. That path is implemented, but a project-permission error on the Earth Engine account blocked it during this study, so no reported result uses GEDI.

D. Clouds

Southeast Asia is cloudy on at least 80% of days, with no reliable clear season [48], and Sentinel-2 metadata for 2017 showed at most one cloud-free scene over much of tropical Indonesia [49]. Tropical maps of clear-observation frequency exist at 30 m [77], but we found no Sentinel-2 statistic at Indonesian plot scale and no measurement of how often scene-level filters misjudge a plot, so we measured both, then designed the estimator so that clouds cost as little as possible.

C1, plot-level climatology. For 97 locations (one per 0.1° cell holding a plot), we read the Sentinel-2 scene classification (SCL) of every acquisition from 2021 to 2023 in a 5×5 window of 20 m pixels around the plot: 22,697 acquisitions after duplicates of the same date were merged. An acquisition is *clear* if every pixel is vegetation, bare soil, water, or unclassified. From this we estimate the per-acquisition clear probability p, its monthly cycle, the gaps between clear looks, and the probability of at least one clear look within $\pm W$ days of a random date, which we compare with the independence model $1 - (1 - p)^m$ for m acquisitions in the window; the difference measures how much clouds cluster in time. We also test the scene-level filter used by many Earth Engine pipelines, including our own platform (`CLOUDY_PIXEL_PERCENTAGE` < 30 or < 40), against the plot condition.

C2, missing optical data. When clouds hide a plot, the optical block (Sentinel-2 and Landsat features) is missing. For a fraction $q \in \{0, 0.25, 0.5, 0.75, 1\}$ of each target region's plots we remove it and compare four strategies, all leave-one-region-out: *naive* (the full model, missing values routed down LightGBM's default branches); *dropout*, one model trained on the real rows plus copies with the optical block removed, the modality-dropout idea of Neverova *et al.* [58]; *switch*, a cloud-aware ensemble that uses a radar-and-terrain model when optical data are missing; and *impute*, optical features regressed from radar, a simple form of SAR-to-optical gap filling [57].

Cloud-aware conformal intervals. A prediction interval is only honest if it is calibrated on cases like the one being predicted. When clouds are rare in the calibration data, a group-conditional (Mondrian) calibration [62] of the cloud-hit group has too few scores. We therefore use the missing-data augmentation of Zaffran *et al.* [72]: every inner out-of-fold calibration plot is scored twice, as observed and with its optical block removed, with quantile models trained under modality dropout [58], and each test plot takes the threshold of its own missingness pattern. With two patterns this is a special case of their CP-MDA. Their guarantee assumes exchangeable data and missingness independent of the outcome; here clouds follow climate and region, and calibration is out-of-region, so neither holds and coverage has to be measured. We test it on every target plot, both as observed and fully clouded.

E. Estimator Ladder

E0 (off the shelf). Global products read at the plot, no fitting. **E1 (no field data).** LightGBM trained only on GEDI footprint biomass labels sampled around each region, using the same satellite predictors; no field plot enters training (implemented, not yet evaluated). **E2 (transfer).** LightGBM on $\ln$ AGB, trained on all region blocks except the target block and predicting it blind (leave-one-region-out), with Duan smearing [22]. **E3 (k-shot).** E2 followed by a ratio calibration on k randomly drawn plots of the target block, scored on its remaining plots, repeated 200 times per k. Split-conformal intervals for E2 are calibrated on inner leave-one-region-out residuals of the training blocks only [25].

F. Confidence: Plot and Project Intervals

Plot intervals. All intervals target 90% and are calibrated on inner leave-one-region-out residuals of the training regions only. With calibration scores $s_1, \ldots, s_n$, the conformal threshold is the $\lceil (n+1)(1-\alpha) \rceil$-th smallest score, which guarantees marginal coverage between $1-\alpha$ and $1-\alpha+1/(n+1)$ under exchangeability [59]. We compare four scores on $y = \ln \mathrm{AGB}$:

$$\text{split:} \quad s = |y - \hat{y}|, \qquad \hat{C} = \hat{y} \pm q, \tag{7}$$

$$\text{Mondrian:} \quad \text{split, one } q_g \text{ per availability group } g, \tag{8}$$

$$\text{normalized:} \quad s = |y - \hat{y}|/\hat{\sigma}(x), \qquad \hat{C} = \hat{y} \pm q\,\hat{\sigma}(x), \tag{9}$$

$$\text{CQR:} \quad s = \max\{\hat{q}_{\mathrm{lo}}(x) - y,\ y - \hat{q}_{\mathrm{hi}}(x)\}, \tag{10}$$

$$\hat{C} = [\hat{q}_{\mathrm{lo}}(x) - q,\ \hat{q}_{\mathrm{hi}}(x) + q], \tag{11}$$

following Papadopoulos *et al.* [61] for the normalized score, with $\hat{\sigma}(x)$ a second LightGBM fitted to the inner residuals,

and Romano *et al.* [60] for conformalized quantile regression (CQR), with LightGBM quantile models at 5% and 95%. We report coverage, the median ratio of the upper to the lower bound, and the Winkler interval score on the log scale, $(\hat{u} - \hat{l}) + \frac{2}{\alpha}\big[(\hat{l} - y)_+ + (y - \hat{u})_+\big]$. As Araza *et al.* note, an interval calibrated on plots covers the plot value, which includes field error, not the true stock [67].

Is a metadata "confidence score" a confidence? Monitoring platforms often grade their output with a score built from cloud cover, data completeness, and temporal offset. Ours did: 40 points for cloud cover, plus weights for completeness and stability. We rebuild such a score from the same inputs (40 points for clear optical scenes, 30 for the share of sensor groups present, 30 for temporal closeness) and test whether it ranks plots by their actual error, using Spearman's ρ against $|\ln y - \ln \hat{y}|$, next to the learned $\hat{\sigma}(x)$, both raw and as a partial rank correlation that removes predicted AGB and the dataset or region, which could otherwise confound it.

Project intervals. Every region block with at least eight plots plays a project, whose true stock is the mean field AGB of its plots. For satellite-only estimates of the mean, a naive interval treats plot errors as independent, $SE = s_w/\sqrt{n}$. Map errors, however, are spatially correlated over 50–100 km [67], so a project shares one regional bias. The two-component interval adds the between-region variance of the region-mean log residual,

$$SE^2 = \frac{s_w^2}{n} + s_b^2, \tag{12}$$

with s_w and s_b estimated from the other regions only. When k field plots are available, the satellite becomes auxiliary information for a model-assisted estimator [70], [71]. The ratio estimator is

$$\hat{\bar{Y}}_R = \frac{\sum_{i\in s} y_i}{\sum_{i\in s} \hat{y}_i}\,\bar{\hat{y}}, \qquad \widehat{V} = \Big(1 - \frac{k}{n}\Big)\frac{1}{k}\,s_e^2, \quad e_i = y_i - \hat{R}\hat{y}_i, \tag{13}$$

where $\bar{\hat{y}}$ is the satellite mean over the whole project. The regression (GREG) estimator is $\bar{y}_s + \hat{\beta}(\bar{\hat{y}} - \bar{\hat{y}}_s)$. Both are compared with the field-only mean of the same k plots over 500 random draws. The relative efficiency $\mathrm{MSE}_{\text{field}}/\mathrm{MSE}_{\text{estimator}}$ says how many field plots one satellite-assisted plot is worth. We use the empirical MSE over draws, because the GREG variance formula underestimates at small k. A conservative estimate $\hat{Y}(1-\mathrm{UNC})$ with the VM0047 rule exceeds the truth whenever the project is over-credited; we report how often.

G. Scoring

We report RMSE, relative RMSE, bias, R^2, and Lin's concordance [26] per dataset and overall, interval coverage for E2, error by field-AGB range, error versus window W, and the error of the mean of n plots as n grows, which approximates estimating a larger area.

H. Sequestration

Credits are issued on change, not stock. The only public tropical dataset we found that gives repeated per-stem measurements with *day-precision* dates is the SAFE Project tree census in Sabah, Malaysian Borneo [7], [8]: 25 m×25 m plots across old-growth, logged, riparian, and oil-palm land, censused from 2010 to 2020, with every stem's measurement date recorded. It is also the closest field site to Kalimantan in our benchmark.

The right grid applies the same chain to every census: Chave Eq. 4 on stems with $D \geq 10$ cm; height from a local $\ln H = a + b \ln D$ model fitted to the heights SAFE measured, with Duan correction; wood density from the Global Wood Density Database [9], [10] by species, then genus, then family, then plot mean, preferring South-East Asian records. For each plot with censuses at least three years apart, sequestration is

$$\Delta\mathrm{AGB}_j = \frac{\mathrm{AGB}_j(t_1) - \mathrm{AGB}_j(t_0)}{t_1 - t_0} \ \ [\mathrm{Mg\,ha^{-1}\,yr^{-1}}], \tag{14}$$

where t_0 and t_1 are the median stem dates of the first and last census. This gives 223 plots in 18 blocks, with t_0 between 2010 and 2015 and t_1 between 2016 and 2020. The manifest released to the left grid holds only plot id, block, coordinates, and the two dates.

The left grid estimates ΔAGB three ways: *no change* ($\Delta = 0$, the baseline any estimator must beat); *S0*, the difference of the Chloris annual maps between the census years; and *S2*, the difference of two E2 stock predictions at t_0 and t_1, with E2 trained on the 476 stock plots only. Sentinel-2 did not exist at most t_0 dates, so S2 uses Landsat, Sentinel-1, PALSAR, and terrain. No SAFE label is released to any estimator; the same E2 model also predicts SAFE stock at t_0, which is an out-of-region test on Borneo.

I. Blindness Ledger

Every protocol step is an entry in an append-only log whose entries carry the SHA-256 of the previous entry [27]. An evaluation epoch is: `seal` (digest of the sealed outcomes and of the public manifest), `commit` (digest of each estimator's prediction file), `release` (for E2, the digest and size of each fold's training labels, with a flag stating whether any target-block label was included), and `reveal` (accepted only if the sealed file still matches its digest). Commits after a reveal are refused. Editing any entry breaks every later hash, so a third party can verify the order of events from the log alone [28].

J. Distributed Extraction

One task is one (plot, date, window, sensor) request. Its result is stored under the SHA-256 of the canonical task, which makes re-execution idempotent and turns the cache into a checkpoint: a restarted run skips every finished task. Writes are atomic (write then rename). Transient errors are retried with capped exponential backoff and jitter. Tasks are grouped into spatial tiles so a worker reuses scenes for neighbouring plots, but field plots cluster (one site holds 89 plots), so a tile holding more than the per-worker mean is split into sub-buckets by plot id, keeping all sensors of a plot together. Buckets are then placed either by consistent hashing [30] or by the longest-processing-time rule [31]. On the 2,052 tasks of this study with 24 workers, the worst-worker load factor $\lambda = \max_w L_w/\bar{L}$ was 6.75 for tile-only placement, 2.18 for

TABLE III
RIGHT-GRID UNCERTAINTY PER PROVINCE (HCSA, AGB)

			u_{90} (%)		Meas.	VM0047	Plots for
Province	n	$\bar{b}$	**Sampl.**	**Total**	**share**	**deduct.**	**±10%**
C. Kalimantan	33	250	13.8	14.0	10%	3.8%	36
Jambi	20	171	33.9	34.5	16%	23.9%	71
Riau	19	126	12.7	19.8	64%	2.7%	19
S. Sumatra	22	80	9.2	12.3	51%	0.0%	14
W. Kalimantan	36	165	14.4	15.1	15%	4.4%	23
W. Papua	18	116	8.8	12.4	57%	0.0%	8

$\bar{b}$: stratified mean AGB (Mg ha^{-1}). Meas. share: measurement and allometry share of the variance of $\bar{b}$.

TABLE IV
BLIND STOCK ACCURACY (MG AGB HA^{-1}; COMMITTED BEFORE REVEAL)

Subset	**Estimator**	**RMSE**	**rRMSE**	**Bias**	**R^2**	**CCC**
All (476)	E0 Chloris	128.1	49.3%	−4.6%	0.29	0.43
	E0 HGB 2010	149.4	57.5%	−8.4%	0.03	0.27
	E2 LORO	126.1	48.5%	−9.5%	0.31	0.59
GEO-T. (142)	E0 Chloris	102.8	37.5%	−10.3%	0.37	0.56
	E2 LORO	114.9	41.9%	0.0%	0.21	0.55
HCSA (148)	E0 Chloris	172.4	107.1%	+19.0%	−0.12	0.02
	E2 LORO	165.6	102.9%	−29.1%	−0.03	0.22
Labr. (186)	E0 Chloris	101.2	31.0%	−10.3%	0.18	0.37
	E2 LORO	93.7	28.6%	−7.9%	0.30	0.55

split tiles with consistent hashing, and 1.40 for split tiles with LPT.

IV. RESULTS

A. How Certain Is the Right Grid?

Before scoring the satellites we score the field answer.

Plot level. On the 148 HCSA plots, the Monte Carlo 95% interval of plot AGB has a median half-width of 31% (IQR 25–45%). Of the stems, 45% have a species-level wood density, 42% genus-level, 7% family-level, and 5% none. The Monte Carlo median is 5% above the deterministic value, because the allometric error is log-normal; the deterministic value is the one the standards use.

Equation choice. Consistent with earlier comparisons of generic and regional equations [74], the regional equation of Basuki *et al.* gives almost the same total over all plots (ratio 1.01). By region, however, the stock moves from −40% (Central Kalimantan) to +121% (South Sumatra), because it ignores height where HCSA measured short trees.

Pool choice. The root-to-shoot ratio of SNI 7724 (IPCC 2006) gives 13% more total carbon than the 2019 Refinement. This is close to the 12% shift in global vegetation carbon that Mokany *et al.* found from revising root-to-shoot ratios [75]. Two standards, both valid, differ by more than the precision target of most methodologies.

Project level. Table III applies AR-TOOL14 to each province. The sampling uncertainty at 90% is 9–34%, and five of six provinces meet SNI 7724's 20% limit. But measurement and allometric error, which AR-TOOL14 does not count, carry 10–64% of the variance of the project mean. Their share is largest where the plots vary least between each other (Riau, 64%). The deductions differ by standard for the same data (Jambi: 24% under VM0047, 34% under AR-TOOL14, 14% under Gold Standard). Reaching ±10% at 90% would take 8 to 71 plots per province.

Change. Growth is far harder than stock. On SAFE, the Monte Carlo SD of a plot's ΔAGB is 2.0 Mg ha^{-1} yr^{-1} when a tree keeps its allometric residual across censuses, against 4.3 when it is redrawn. The signal itself is 1.9. Remeasuring the same stems with the same equation cancels half the error. At block level, Eq. 6 gives 90% uncertainties of 101–901% when the correlation of the two stock means is ignored ($\rho = 0$), which is a 100% deduction for 13 of 15 blocks and over 90% for the other two. With the measured ρ (0.45–0.99), the uncertainty falls to 28–469%, and seven blocks keep some credit, the best at an 18% deduction. With 9–16 plots of 0.06 ha per block, even the field standard cannot certify growth of 2 Mg ha^{-1} yr^{-1}, unless it uses the correlation that permanent plots provide.

B. Blind Accuracy of Carbon Stock (RQ1)

Table IV gives the blind scores. All three estimators were committed to the ledger before the reveal. Over all 476 plots the best blind estimate, E2, misses the field value by a relative RMSE of 48.5% and explains 31% of its variance; the Chloris product, which was never fitted to any of our plots, is almost as close (49.3%). The 2010 harmonized map is worse (57.5%, $R^2 = 0.03$), partly because it describes 2010 while most plots were measured years later. Because carbon is a fixed multiple of AGB on both grids, every relative figure applies unchanged to t C ha^{-1} and t CO_2e ha^{-1}.

The dataset matters more than the estimator. On the 1 ha Labrière units E2 is within 28.6% (CCC 0.55); on GEO-TREES within 41.9%; on the Indonesian HCSA plots every estimator misses by more than 100% and none beats predicting the mean ($R^2 < 0$).

C. Where the Error Comes From

Saturation. Fig. 2 shows the pattern. Binned by field AGB, both E0 and E2 regress toward ∼300 Mg ha^{-1}. For plots above 400 Mg ha^{-1} (mean 480) the E2 mean prediction is 293 and the Chloris mean is 295, a median under-estimate of about one third. Below 50 Mg ha^{-1} (mean 21) E2 predicts 60 and Chloris 178. Between 100 and 300 Mg ha^{-1} the median error of E2 is only +5 to +12%. The satellite signal therefore carries useful information in the middle of the range and little at either end, which is the behaviour reported for optical and L-band signals [18], [19].

Scale. The HCSA plots are nested 0.01/0.05 ha plots with a median of 15 stems, so one large tree more or less moves the plot by hundreds of Mg ha^{-1}. Their field values span 2 to 876 Mg ha^{-1} within one province. A 30 m pixel window cannot resolve that, and the 4.6 km Chloris pixel averages

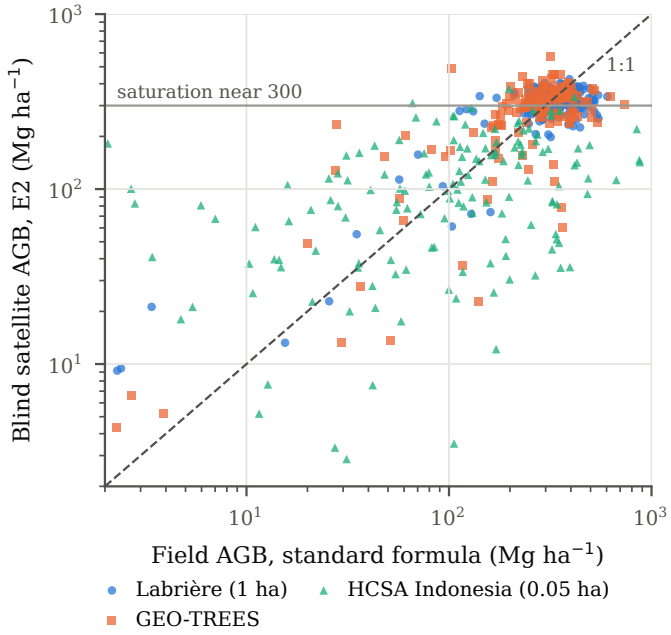


Fig. 2. Blind E2 prediction against the field value for 476 plots (log axes). Above about 300 Mg ha^{-1} predictions flatten; small HCSA plots scatter widely in both directions.

TABLE V
IMPROVEMENT LADDER, BLIND RRMSE (%)

Left-grid inputs	All	GEO-T.	HCSA	Labr.
Optical (S2 + Landsat)	56.9	51.2	103.0	42.0
Radar (S1 + PALSAR)	54.5	45.9	114.7	33.7
Optical + radar	52.1	46.4	103.7	33.9
+ terrain (= E2)	49.1	42.0	104.8	28.8
+ Chloris as feature	47.9	38.1	105.7	27.9
+ plot footprint	47.6	38.6	103.5	28.1
Mean of E2 and Chloris	**45.4**	**35.0**	**98.8**	28.0

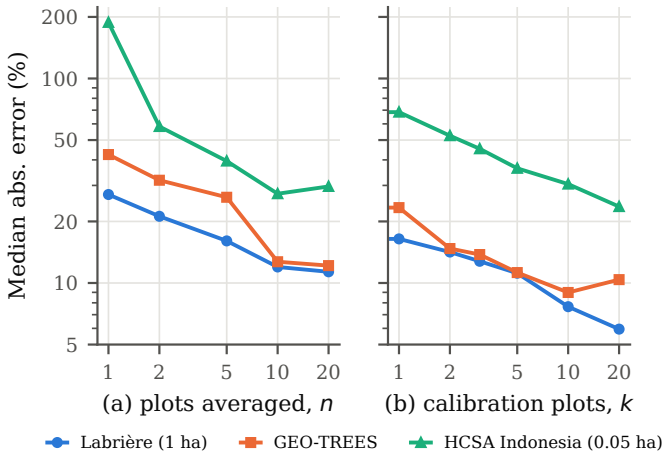


Fig. 3. Closing the gap. (a) Error of the mean of n plots against its field mean; (b) error of the block total after a ratio calibration on k local plots. Median over blocks and 200–300 random draws; log axes.

open land and forest together, which explains its large overestimate on HCSA open-land and scrub plots. The HCSA land-cover labels would separate these plots well (within-class standard deviations of 10–40 Mg ha^{-1} for all classes except high-density forest), but they are released with the field records, so they belong to the right grid and are never shown to the left grid. A satellite classifier that reproduced the HCS stratification is a promising left-grid estimator for future work.

Field truth is not the bottleneck. Labrière publish a 95% credible interval for every unit; the implied standard error of the field value is 5.0% (RMS), against a 28.6% satellite error on the same units. The right grid is far more precise than the left grid, so the gap is a satellite gap, not a ground-truth gap.

Time. Re-extracting the Indonesian plots' optical and C-band features with $W = 30$, 90, 180, and 365 days changed their blind error by less than three points (rRMSE 102.5, 104.8, 102.7, 102.0%), while the share of plots with at least one clear Sentinel-2 scene rose from 74% at 30 days to 92% at 90 and 100% at 180. A tighter window costs coverage in the tropics and buys no accuracy when the field date is only known to the year, and when plot-scale noise dwarfs seasonal change. Temporal matching matters where the forest changes between the two dates, which is the sequestration case below.

D. Closing the Gap (RQ2)

Better models. Table V adds one information source at a time, always leave-one-region-out. These variants were run after the reveal, so they are exploratory; every variant we tried is reported. Radar alone is ahead of optical alone (54.5% vs 56.9% overall; 33.7% vs 42.0% on the 1 ha units), the two together beat either, terrain helps further, and feeding the Chloris product to the model as one more feature helps again. Of these steps, only the last one is clearly outside sampling noise (Section IV-G): the simple average of E2 and Chloris is the best blind estimator overall (45.4%, $R^2 = 0.39$), 3.7 points better than E2 with a region-bootstrap 95% interval of -7.2 to -0.6. None of these steps rescues the HCSA plots.

A few local plots. The largest gain comes not from the model but from field data in the target region. Fig. 3(b) shows E3: with k plots of the target block used for a ratio correction, the error of the block total falls from 29% (GEO-TREES) and 11% (Labrière) at $k = 0$ to 9–10% and 6% at $k = 10$–20 (median over blocks and 200 draws). On HCSA, calibration hurts until $k = 20$ (27% at $k = 0$, 53% at $k = 2$, 24% at $k = 20$): a small plot is so noisy that a handful of them is a worse calibrator than none. How many local plots are needed depends on plot size as much as on the satellite.

Aggregation. Credits are issued for project areas, not single plots. The error of the mean of n plots shrinks with n until only systematic bias remains (Fig. 3(a)). For the 1 ha units the median absolute error falls from 27% for one plot to 12% for ten and 11% for twenty; for HCSA from 188% to 27–30%. The floor of roughly 10% outside Indonesia and 25–30% in Indonesia is the regional bias that only local calibration removes.

TABLE VI
PLOT-LEVEL SENTINEL-2 CLEAR-SKY CLIMATOLOGY, 2021–2023 (MEDIAN OVER LOCATIONS)

			Gap (d)		P(clear in $\pm W$)	
Region	**Loc.**	p	**Med.**	**P90**	± 15 d	± 30 d
Indonesia (HCSA)	24	0.215	10	40	0.81	0.96
Sabah (SAFE)	10	0.262	15	41	0.84	0.99
Africa	22	0.166	18	62	0.61	0.79
Americas	40	0.394	10	25	0.82	0.94

p: share of acquisitions with every plot pixel clear. Independence model at ± 30 d: Indonesia 0.98, Sabah 0.98, Africa 0.89, Americas 1.00.

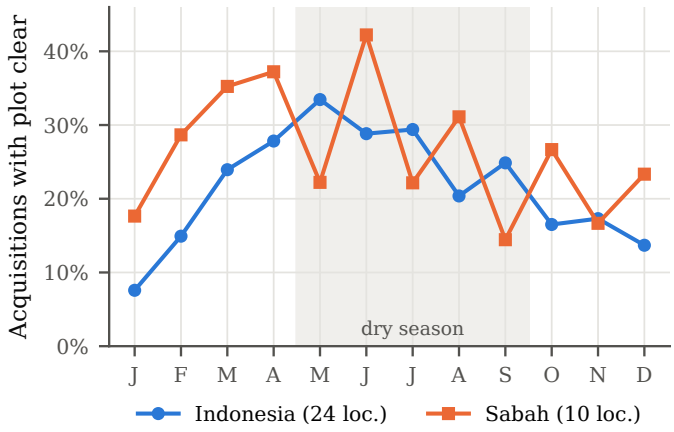


Fig. 4. Share of Sentinel-2 acquisitions in which the whole plot is clear, by month, 2021–2023.

Intervals are analysed together with clouds in Section IV-E.

E. Clouds and Confidence (RQ3)

How often is a plot clear? Table VI gives the plot-level climatology. On a median Indonesian HCSA location, only 21.5% of Sentinel-2 acquisitions see the plot completely clear (Sabah 26%, Gabon and Congo 17%, the Americas 39%). The year is not flat (Fig. 4): in Indonesia the clear share falls to 7.6% in January and 14–15% in December and February, and peaks at 33.5% in May, a factor of 4.4. Revisit compensates for cloudiness. With two Sentinel-2 satellites, the median gap between clear looks is 10 days and the 90th percentile 40 days, so 81% of random dates have a clear look within ± 15 days and 96% within ± 30. Clouds cluster in time: the measured probability falls below the independence model in every region, most in Africa (± 30 days: 79% against 89%).

The scene filter is the real problem. Scene-level cloud percentage agrees only loosely with the plot (Spearman -0.67). In Indonesia, the rule `CLOUDY_PIXEL_PERCENTAGE` < 30 keeps 11.5% of acquisitions, but 34% of the scenes it keeps are cloudy over the plot, and it throws away 65% of the acquisitions in which the plot is clear. At < 40, the threshold our own platform uses for segmentation, 40% of the kept scenes are cloudy over the plot. Masking at the plot instead of filtering scenes gives 2.8 times as many usable clear looks (21.5% against 7.6% of acquisitions) and removes the cloudy ones that pass the filter.

TABLE VII
BLIND RRMSE (%) WHEN CLOUDS REMOVE THE OPTICAL BLOCK FROM A SHARE q OF TARGET PLOTS

q	**Naive**	**Dropout**	**Switch**	**Impute**
0	48.0	48.0	**47.7**	51.5
0.25	48.5	48.6	**47.9**	61.2
0.5	49.6	49.1	**49.1**	67.9
0.75	49.3	49.4	**49.1**	77.5
1	**50.0**	**50.0**	50.4	84.6

TABLE VIII
90% INTERVAL METHODS, LEAVE-ONE-REGION-OUT (COVERAGE; MEDIAN UPPER/LOWER RATIO; WINKLER SCORE)

Method	**All**	**GEO-T.**	**HCSA**	**Labr.**	**Ratio**	**Winkler**
Split	0.884	0.958	0.682	0.989	16.3	4.21
Mondrian	0.882	0.958	0.676	0.989	16.3	4.42
Normalized	0.794	0.782	0.723	0.859	3.3	3.53
CQR	**0.899**	0.937	**0.811**	0.941	5.3	**3.05**

What clouds cost the estimate. Table VII removes the optical block from a growing share of target plots. Because the model leans on radar, terrain, and the annual product, losing every optical feature raises the blind error from 48.0% to 50.0% rRMSE. Modality dropout and the cloud-aware switch both handle partial cloud well; the switch is best on the cloud-hit plots at low q (37.7% against 43.3% for the naive model at $q = 0$, on the 23 plots that really had no clear scene). Imputing optical features from radar is harmful: the bias grows to -61% at full cloud, a warning for SAR-to-optical gap filling used as a model input.

Which intervals are honest? Table VIII compares the four interval methods. Split conformal, the method of the confirmatory run, reaches 88% coverage but is wide (upper bound 16 times the lower) and fails on HCSA (68%). Normalized conformal is narrow but under-covers (79%), because a difficulty model fitted on other regions does not transfer. CQR is the only method that reaches the target overall (89.9%), with intervals three times narrower than split (ratio 5.3) and the best Winkler score. Mondrian calibration by optical availability adds nothing: only 23 plots lack optical data, too few to calibrate a group. HCSA remains under-covered by every method (68–81%): its plots are noisier than any training block, a shift that marginal conformal guarantees do not cover.

Intervals under cloud. Fig. 5 shows the effect. When every target plot is clouded, plain CQR, calibrated on plots as observed, falls to 86.5% coverage (HCSA 70%). The cloud-aware CQR, calibrated on synthetic cloud, keeps 90.5% (HCSA 77%, GEO-TREES 95%, Labrière 98%). It does this by widening the interval honestly, to a ratio of 7.8 against 5.4. On plots as observed, it covers 90.5% at ratio 5.7. A cloudy plot gets a wider interval, which is what an honest confidence statement should do.

A metadata score is not a confidence. Over all plots,

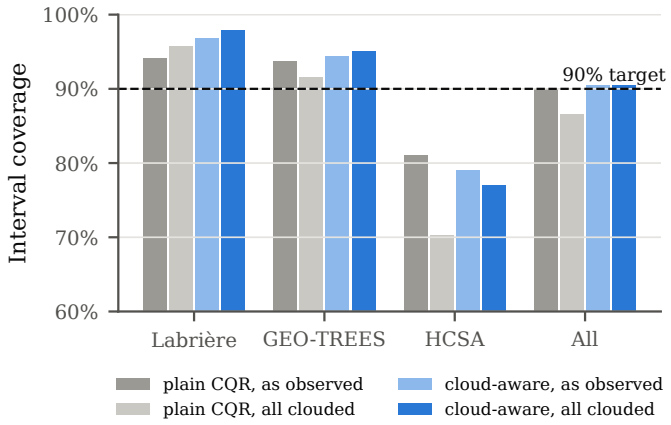


Fig. 5. Coverage of 90% intervals on target plots as observed and with every optical feature removed. Cloud-aware calibration keeps the target overall; plain CQR falls to 86.5%.

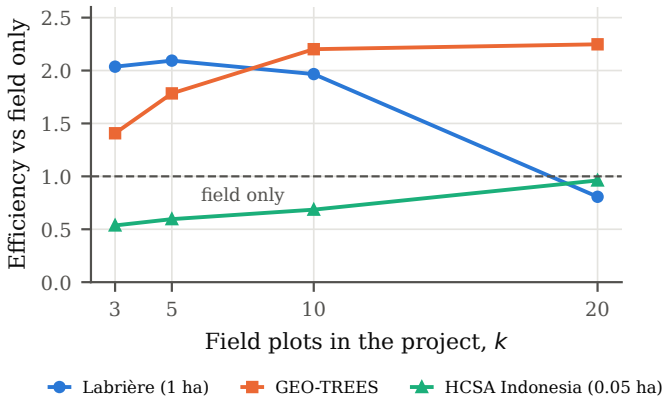


Fig. 6. Efficiency of the ratio estimator (satellite + k field plots) relative to the field-only mean, by empirical MSE over 500 draws. Above 1, the satellite saves field plots.

the rebuilt platform-style confidence score correlates with the actual error in the *wrong* direction (Spearman $+0.39$): plots scored as most reliable were the least accurate. That figure is confounded. The recent, cloud-checked HCSA plots score high and are the hardest to predict. Once predicted AGB and the dataset (or the region) are removed, the partial rank correlation is -0.03 to $+0.07$, so the score carries no information about error beyond which dataset a plot belongs to. The learned difficulty $\hat{\sigma}(x)$ fares no better at plot level (partial correlation $+0.02$ to $+0.07$). Neither a metadata score nor a learned difficulty can rank individual plots by error here. Confidence therefore has to be stated as a calibrated interval and checked against field data, as found for global biomass maps [67], [83].

Project-level confidence and over-crediting. For 19 projects (blocks with at least 8 plots), the blind satellite mean missed the field mean by a median of 23%. The naive interval, with plot errors assumed independent, had a median half-width of 28% but covered only 74% of projects (HCSA 33%), and two projects would have been over-credited even after the VM0047 deduction. The two-component interval of Eq. 12 covered all 19, at an honest half-width of 71–86%, which under VM0047 means 61–76% of the credits are deducted. This agrees with earlier coverage studies of model-based estimators that omit residual spatial correlation or model bias [81], [82]; what our test adds is the size of the effect when a model is transferred across regions. A satellite-only project estimate is therefore either over-confident or almost uncreditable.

Field plots change this (Fig. 6, Table IX). With the satellite as auxiliary data, the ratio estimator is 1.4 to 2.3 times as efficient as the field-only mean on GEO-TREES ($k = 3$ to 20) and about 2 times on Labrière ($k = 3$ to 10); at $k = 20$ the small Labrière regions are nearly fully sampled and the advantage disappears. Equivalently, the satellite replaces roughly half of the field plots needed for a given precision: summed over regions, the plots needed for $\pm 10\%$ at 90% fall from 408 to 216 (GEO-TREES) and from 381 to 162 (Labrière). On HCSA, where the satellite cannot rank small plots (within-region correlation 0.2), the ratio estimator is less efficient than field-only at every k (0.54–0.96), only GREG gains at $k = 20$ (1.25), and the plots needed rise from 1,483 to 1,877. The gain grows with plot size, as found for airborne lidar and InSAR in Tanzania [80]. GREG under-states its own variance at small k (coverage 66–73% at $k = 3$), so the ratio estimator is the safer choice. Finally, the VM0047 deduction does not make an estimate conservative: with $k = 10$ plots, the deducted estimate still exceeded the true stock in 32–43% of draws on the large-plot datasets, for all three estimators, because the deduction only starts above 10% uncertainty. Analytically, the deducted estimate is $\hat{C}(1 - U + 0.10)$, ten points of $\hat{C}$ above the one-sided 95% lower bound, so its exceedance probability must lie between about 5% and 50%; the simulation places it for realistic designs. Critiques of registry deductions so far have been qualitative [84], [85], and over-crediting has been documented for other rules [86].

TABLE IX
SATELLITE AS AUXILIARY DATA: COVERAGE OF 90% CI AND EFFICIENCY VS FIELD-ONLY (500 DRAWS)

		Field only		Ratio (satellite + k)		
Dataset	k	**Cov.**	**RMSE**	**Cov.**	**RMSE**	**Eff.**
GEO-TREES	5	0.91	16.3	0.90	12.2	1.78
	10	0.88	10.4	0.89	7.0	2.20
	20	0.90	5.7	0.90	3.8	2.25
HCSA	5	0.85	36.7	0.80	47.6	0.60
	10	0.86	22.4	0.86	27.1	0.69
	20	0.89	11.1	0.87	11.3	0.96
Labrière	5	0.89	17.9	0.86	12.3	2.09
	10	0.88	10.1	0.88	7.2	1.97
	20	0.88	3.1	0.89	3.5	0.81

RMSE in % of the project mean. Eff.: MSE(field)/MSE(ratio); > 1 saves plots.

F. Sequestration (RQ4)

A right-grid failure first. After the reveal, the 27 oil-palm plots showed a mean field stock of 505 Mg ha^{-1}, more

TABLE X
BLIND SEQUESTRATION, SAFE EXCLUDING OIL PALM (MG AGB HA^{-1} YR^{-1}; FIELD MEAN 1.87)

	Plot (196)			**Block mean** (15)	
Estimator	**RMSE**	**Bias**	r	**MAE**	**Bias**
No change	8.12	−1.87	–	2.05	−1.92
S0 Chloris difference	8.06	−1.01	0.07	1.61	−1.20
S2 E2 stock difference	16.36	−5.90	−0.02	7.40	−7.08
S2c consistent sensors†	13.96	−3.50	0.09	4.48	−3.77
S3 change model, LOBO†	8.39	−0.50	−0.07	1.33	−0.27
Regional default rate†	**7.91**	**0.00**	–	**0.98**	−0.05

†Exploratory, defined after the reveal.

TABLE XI
RELIABILITY: REGION-BLOCK BOOTSTRAP 95% INTERVALS (2,000 DRAWS) AND SEED SPREAD

Quantity	**Value**	**95% interval**
rRMSE E2, all plots	48.5%	34.9–68.7
rRMSE E2, Labrière (1 ha)	28.6%	22.8–38.4
rRMSE E2, GEO-TREES	41.9%	35.9–51.8
rRMSE E2, HCSA	102.9%	88.1–121.7
E2 − Chloris (points)	−0.8	−8.3 to +5.3
E2–Chloris average − E2 (points)	−3.7	−7.2 to −0.6
Radar only − optical only (points)	−2.3	−11.1 to +5.1
CQR coverage, as observed	89.9%	83.9–94.4
Cloud-aware CQR coverage, fully clouded	90.5%	84.7–95.2
Cloud-aware − plain, fully clouded (points)	+4.0	+1.9 to +6.6
E2 rRMSE over 5 seeds (mean ± SD)	48.8	±0.36

than old-growth forest. Every stem in those plots is an oil palm whose recorded "diameter" (median 76 cm) includes the frond bases, and Chave Eq. 4 is a model for dicotyledonous trees. The standard formula is therefore not valid there. The evaluation script had fixed an "excluding oil palm" subset before the reveal, and we report it as the primary result: 196 plots in 15 blocks, mean field $\Delta\mathrm{AGB} = 1.87\,\mathrm{Mg\,ha^{-1}\,yr^{-1}}$ ($0.88\,\mathrm{t\,C\,ha^{-1}\,yr^{-1}}$), median census gap 8.4 years.

Blind satellite change estimates do not beat "no change". Table X gives the scores. At plot level the Chloris difference (S0) has almost the same error as predicting zero (RMSE 8.06 vs $8.12\,\mathrm{Mg\,ha^{-1}\,yr^{-1}}$, $r = 0.07$). The E2 stock difference (S2) is twice as bad and has the wrong sign: it predicts a loss of $4\,\mathrm{Mg\,ha^{-1}\,yr^{-1}}$ where the forest gained 1.9. The cause is a sensor artefact: Sentinel-1 existed for only 13% of the plots at t_0 but for all of them at t_1, and no L-band mosaic before 2015 is available in the catalogue, so the two stock predictions are made from different inputs. Restricting the stock model to sensors present at both dates (S2c: Landsat and terrain) halves the bias but not the noise.

Why. The blind E2 stock error on these plots has a standard deviation of $146\,\mathrm{Mg\,ha^{-1}}$. If the errors at t_0 and t_1 were independent, differencing them over 8.4 years would add noise of about $25\,\mathrm{Mg\,ha^{-1}\,yr^{-1}}$, thirteen times the signal. The errors are partly correlated, because the same place is seen twice, which is why the observed S2c error ($14\,\mathrm{Mg\,ha^{-1}\,yr^{-1}}$) is lower, but it is still seven times the signal. A stock estimator with plot-scale error of 30–100% cannot resolve growth of 1% per year.

What does work is not a satellite. Two estimators trained on the other SAFE blocks' measured change (leave-one-block-out, run after the reveal and therefore exploratory) were added: a LightGBM on Landsat and Chloris change features (S3), and the plain mean rate of the other blocks with no satellite input at all. The regional default rate is the most accurate estimator at both scales (block-mean MAE 0.98 vs 2.05 for no change), and S3 learns little more than that rate. Such a rate is in effect a local Tier 1 removal factor of the kind compiled by Requena Suarez *et al.* [87]. At this plot size and with these sensors, the information about growth comes from nearby field remeasurements, not from the pixels.

Out-of-region stock on Borneo. The same E2 model, trained without any Bornean plot, misses SAFE stock at t_0 by 132% rRMSE with a +80% bias (excluding oil palm). It over-predicts the 0.06 ha logged and riparian plots (field mean 99, predicted $217\,\mathrm{Mg\,ha^{-1}}$), a second instance of the small-plot problem seen on HCSA.

G. How Reliable Are These Numbers?

Every score above comes from 27 region blocks, and errors are correlated within a block. We therefore resampled whole blocks with replacement (2,000 draws) and refitted E2 with five LightGBM seeds (Table XI). Seed noise is small: the overall rRMSE moved by 0.36 points (SD). Region sampling is not: the 95% interval of the overall rRMSE runs from 35% to 69%. Three kinds of claim survive this test. The first is the differences between datasets and plot sizes, whose intervals do not overlap. The second is the gain of the E2–Chloris average over E2 (−3.7 points, interval −7.2 to −0.6). The third is the coverage gain of cloud-aware calibration under full cloud (+4.0 points, +1.9 to +6.6). Two claims do not survive. The difference between E2 and Chloris (−0.8 points, −8.3 to +5.3) and between radar and optical inputs (−2.3, −11.1 to +5.1) are within noise. We report them as tendencies, not findings.

H. Systems (RQ5)

Production runs. The stock extraction ran 2,052 tasks on 24 workers in 2.1 h ($0.26\,\mathrm{tasks\,s^{-1}}$) with a measured load factor of 1.47, close to the 1.40 predicted from task counts; 85 transient errors were absorbed by retries. We stopped the first attempt, which used tile-only placement, to switch policies; the restart skipped the 92 tasks already in the content-addressed cache. Seventeen tasks (0.8%) failed permanently: 13 hit two Landsat scenes whose files could not be read from the catalogue, and four hit unreadable Sentinel-2 files. Retrying cannot repair a permanently broken input; the runner records such tasks and the model treats their features as missing, and a scene-level fallback is the obvious next step. The SAFE extraction ran 2,006 tasks on 16 workers in 2.3 h with no permanent failures, 36 retries, and a load factor of 1.44.

Scaling, faults, and resume. A random sample of 48 tasks (Landsat, Sentinel-1, PALSAR, terrain) was run on 1 to 32

TABLE XII
STRONG SCALING AND FAULT INJECTION (48 TASKS, EMPTY CACHE)

Workers	Fault p	Wall (s)	Speed-up	λ	Identical
1	0	498.5	1.00	1.00	47/47
2	0	207.8	2.40	1.00	47/47
4	0	144.8	3.44	1.00	47/47
8	0	98.4	5.06	1.00	47/47
16	0	93.3	5.34	1.33	47/47
32	0	91.4	5.45	2.67	47/47
8	0.1	106.7	4.67	1.00	47/47
8	0.2	105.0	4.75	1.00	47/47
8	0.3	105.7	4.72	1.00	47/47

workers, each time on an empty cache (Table XII). Throughput rose 5.1-fold up to 8 workers and then flattened (5.5-fold at 32). Two limits explain this. With 48 tasks, 32 workers hold one or two buckets each, so the load factor rose to 2.67. And the makespan cannot fall below the slowest single task, a multi-scene read, however many workers wait beside it. The extraction is bound by remote reads, not by local compute. Every run was compared with the single-worker run task by task: all 47 finished results were bit-identical at every worker count. With transient faults injected at rates 0.1, 0.2, and 0.3 per attempt, the runner absorbed 13, 18, and 28 retries, and again all 47 results matched. The one task that never finished, in any run, is the one whose Landsat file is unreadable, which is a permanent fault. In the resume test, a run stopped after 23 tasks; the restart skipped those 23 from the cache, ran the remaining 24, and produced output identical to the uninterrupted run.

The super-linear step from one to two workers reflects variable latency of the remote service between runs; each configuration was run once, so differences of a few seconds are within noise.

Ledger. The final log holds 37 entries : four seals (one per right-grid build and one per evaluation run), three commits, two reveals, and 28 label releases, none of which included a label from its target block. The chain verifies end to end. In unit tests, editing any entry broke verification, a commit after a reveal was refused, and a reveal with a modified outcome file was refused.

V. DISCUSSION

A. Can Satellites Replace the Field Plot?

Not at the scale of a plot, and not for growth. Tested blind, open satellite data put a 1 ha plot's carbon stock within roughly 30–40% of the field-formula value, and a 0.05 ha plot within about 100%; the best honest interval (CQR) is a factor of five wide. For sequestration, the quantity on which credits are paid, no satellite estimator we built beat assuming no change, and a regional average of measured growth beat all of them. A platform that issues plot-level or annual growth numbers from these sensors alone is reporting a default constant, whether or not it says so.

What satellites do well is replace part of the inventory. Used as auxiliary data in a ratio estimator, they were 1.4–2.3 times as efficient as field plots alone on large-plot datasets, roughly halving the plots needed for a given precision (Table IX). The useful framing is not "satellites instead of field plots" but "satellites plus a small, well-designed set of field plots instead of a full inventory". Even that only holds where the satellite can rank plots within a project. On the small Indonesian plots it cannot (within-region correlation 0.2), and there it does not save a single plot.

TABLE XIII
BLIND RESULTS OF THIS STUDY AGAINST PUBLISHED BENCHMARKS

Study	Sensor	Reference	Error
This study, Labrière (blind)	multi-sensor	1 ha plots	28.6%
This study, GEO-TREES (blind)	same	network	41.9%
This study, HCSA (blind)	same	0.05 ha plots	102.9%
Berninger 2018, Kalimantan [68]	S1 + PALSAR	lidar layer	31–38%
Jucker 2018, Sabah [69]	airborne lidar	0.1 / 1 ha	25 / 13%
Araza 2022, CCI [67]	CCI map	0.1° cells	29–61*
Santoro 2021 [15]	GlobBiomass	110 k plots	SD ≈50%

*RMSD in Mg ha^{-1} for AGB below 300; 107 at 300–400. Other errors are rRMSE.

B. Our Numbers Against the Literature

Table XIII places our blind results next to published validations. Our 1 ha error (29%) is close to the 31–38% reported for Sentinel-1 and PALSAR in Kalimantan [68]. That study was validated against a lidar-upscaled reference layer, not blind against field plots, and it reports the same underestimation above 200 Mg ha^{-1}. Our saturation near 300 Mg ha^{-1} matches the flattening near 250 Mg ha^{-1} in the global map of Santoro *et al.* [15] and the steep rise of map error above 300 Mg ha^{-1} found by Araza *et al.* [67]. Airborne lidar in Sabah, the best remote sensing can do, still errs by 25% at 0.1 ha and 13% at 1 ha [69]. Plot size dominates every comparison. Local variation of AGB alone has a coefficient of variation of 46% at 0.1 ha and 17% at 1 ha [47], and Duncanson *et al.* recommend plots of at least 1 ha for validation [66]. If we remove that expected representativeness noise in quadrature, the blind satellite error is about 23% on the 1 ha units and still about 90% on HCSA. Plot size alone does not explain the Indonesian result. The mosaic of open land, scrub, and forest around HCSA plots, and a GPS error comparable to the 12.6 m plot radius [66], are the likely remainder.

C. Clouds: What Actually Matters

Three findings change how clouds should be handled. First, *how often* a plot is clear matters less than *how* clearness is decided. A clear look within ±30 days exists for 96% of dates in Indonesia, because revisit compensates for an 80% cloud frequency [48], [50]. The scene-level filter that most pipelines apply, by contrast, discards two thirds of the clear looks and keeps a third of cloudy ones. Pixel-level masking at the plot, with SCL, Fmask, or Cloud Score+ [51]–[53], nearly triples the usable data. Second, *season matters*: in Indonesia the clear share in May is 4.4 times that in January, so field campaigns and their satellite windows should be planned for the dry months. Third, a model led by radar and an annual product loses only two points of accuracy without any optical

data, so clouds are an availability problem for optical indices but not for a well-designed multi-sensor stock estimate. That is the practical form of the SAR–optical fusion argued for by Reiche *et al.* [56]. Filling optical gaps from radar and feeding them to the model, however, was the worst option we tested. Reconstructed reflectance is a model output whose error does not disappear because the image looks complete.

D. Confidence That Means Something

Honest confidence has three layers, and each failed in a characteristic way. At *plot* level, conformal intervals were honest overall, but only CQR was both honest and usable, and missing-data-augmented calibration [72] kept it honest when optical data were missing, outside the assumptions under which that method is guaranteed. The guarantee is marginal: under the shift from large to small plots (HCSA), coverage fell to 68–81% for every method. Spatially weighted conformal prediction [63], [64] is the next step. At *project* level, independence is the dangerous assumption. Map errors are correlated over tens of kilometres [67], a project shares one regional bias, and an interval that ignores it covered only 74% of projects. The two-component interval is honest, and it shows that a satellite-only project estimate would lose most of its credits under VM0047. At the level of the *standard*, a deduction of $U - 10\%$ is not a lower confidence bound. With ten field plots, the deducted estimate still exceeded the true stock in a third of draws. Crediting the lower bound of a calibrated interval instead would make the over-crediting rate a design parameter rather than an accident.

A metadata quality score answers a different question from a confidence interval. Ours rewarded recent, cloud-checked imagery. Over all plots it trusted most the plots the model got most wrong, and within a dataset it said nothing at all about error. Confidence has to be learned from errors against field data and reported in the units of the claim, as an interval in $\mathrm{t\,C\,ha^{-1}}$, not as a grade.

E. The Ground Truth Is a Model Too

The right grid is fixed by the standards, but the standards leave real choices, and those choices are as large as the precision targets they enforce. Switching the root-to-shoot ratio from the SNI 7724 value (IPCC 2006) to the 2019 Refinement changes total carbon by 13%. Switching from the pantropical Chave equation to the East-Kalimantan equation of Basuki *et al.* leaves the national total unchanged but moves provincial stocks from -40% to $+121\%$. AR-TOOL14 counts sampling error only, yet measurement and allometry carried up to 64% of the variance of a provincial mean. The same data receive deductions from 0 to 34% depending on which standard is applied. For growth the field answer itself is fragile. With 9–16 plots of 0.06 ha, VM0047 Eq. 28 deducts most or all of the credit unless the correlation of permanent plots is used, and the plot-level measurement error of the change halves only when the same stem is assessed with the same equation at both dates. The target "100% accuracy" therefore has no meaning. The meaningful target is to fall within the uncertainty of a field estimate made to a stated standard, and that uncertainty must itself be computed in full.

F. The Path Toward the Field Answer

The experiments rank the levers. (i) *Report at project scale*: aggregation removes most random error. (ii) *Add local plots and use the satellite as auxiliary data*, which is the most effective step, provided the plots are large enough to be ranked by the satellite. (iii) *Mask clouds at the plot, not the scene*, and match the window to the season. (iv) *Lead with structure-sensitive sensors*. Radar was ahead of optical on both large-plot datasets, though within sampling noise, and it is what keeps the estimate alive under cloud; spaceborne lidar (GEDI) and the 2025 L- and P-band missions NISAR and ESA BIOMASS are the next steps; our GEDI-only estimator E1 is implemented but was blocked by an access-permission issue during this study. (v) *Keep sensors consistent through time*: half of the sequestration error came from Sentinel-1 appearing between the two censuses. (vi) *Combine rather than choose*: averaging the transfer model with an independent global product was the best blind stock estimate. (vii) *Calibrate confidence*: CQR with cloud-aware calibration at plot level, the two-component variance at project level, and credit the lower bound.

G. Implications for NusaCarbon

Our audit found default constants where measurements were expected, a scene-level cloud filter, and a confidence grade computed from image metadata. The evidence suggests a concrete redesign:

- Mask clouds per pixel at the plot (SCL or Cloud Score+) instead of `CLOUDY_PIXEL_PERCENTAGE`, and widen the window in the wet season.
- Estimate stock with a radar-led multi-sensor model trained with modality dropout, and attach a cloud-aware CQR interval instead of the letter grade.
- Report projects with the two-component interval, and calibrate each project with ten to twenty large field plots through a ratio estimator.
- Credit the lower confidence bound.
- Credit sequestration from field remeasurement of permanent plots, and use satellites to detect loss and disturbance.
- Log every estimate through the commit-reveal ledger.
- Fix the GEDI band that is read as canopy height.

H. Threats to Validity

Internal.

- The right grid is itself a model: Chave allometry carries its own error, and GEO-TREES and Labrière AGB were computed by their authors, not by us.
- HCSA is released as a 2023 dataset without per-plot dates, and Labrière plots have campaign-year dates only, so temporal matching for those plots is at year precision.
- The standard formula failed on SAFE oil palm. Those plots are excluded by a subset fixed before the reveal, and 45 palm stems remain in forest plots.

- The improvement ladder, the change models, the cloud and interval experiments, and the project-level analysis were defined after the reveals and are exploratory. Only the E0, E2, S0, and S2 scores are confirmatory, and we report every variant we ran; Section IV-G states which differences exceed sampling noise.
- The ledger proves the order of events, not the absence of leakage upstream of it. The Chloris product, used as E0 and as an input of the ensemble, was calibrated by its makers on field and lidar data that we cannot inspect; if any of our plots were among them, that path leaks. E2 without the product is free of this path.
- Provincial stratum weights are plot shares, because mapped stratum areas are not public.
- The cloud climatology uses SCL, whose own overall accuracy is about 80–90% [52].

Construct.

- Plot-scale truth (0.05–1 ha) is compared with 10 m to 4.6 km pixels, and GPS error on small plots adds noise that no estimator can remove.
- Conformal intervals calibrated on plots cover plot values, not true stocks.
- GEDI biomass models were calibrated on field plots elsewhere, so E1 is free of our field data but not of field data altogether.

External. Indonesian plots come from one programme, collected by trained non-foresters; results may not transfer to other forest types or survey teams. Projects in the project-level analysis are region blocks, not registered projects.

VI. Conclusion

We asked whether open satellite data can replace the field plot in forest carbon MRV, and tested it blind. A sealed right grid was computed step by step to SNI 7724, IPCC, CDM, Verra, and Gold Standard rules with a full uncertainty budget. A left grid saw only coordinates and dates. A hash-chained ledger makes any change to a prediction after the answer was opened detectable.

Across 476 stock plots, the best blind estimate missed by 45–49% relative RMSE overall: about 30% on 1 ha plots and about 100% on small Indonesian plots. In Indonesia only 21.5% of Sentinel-2 acquisitions see a plot clear. Masking at the plot instead of filtering scenes nearly triples the usable data, and a radar-led model loses only two points without optical data. Missing-data-augmented conformal calibration kept 90% intervals honest under full cloud, while a metadata confidence score carried no information about error beyond dataset differences. At project scale, an interval that ignores spatially correlated error covered only 74% of projects; with field plots as calibration, the satellite was 1.4–2.3 times as efficient as field plots alone on large-plot datasets. For annual sequestration, no satellite estimator beat "no change", and even the field answer was too uncertain to credit without the correlation of permanent plots.

Satellites are a stock-mapping, disturbance, and plot-saving tool, not a replacement for the field plot, and their confidence must be calibrated, not graded. The contribution is the evaluation, not a new method: the protocol, the ledger, and the extraction runner are reusable. Together they turn a platform's accuracy claim into something a third party can check.

Data and Code Availability

All field sources are public: HCSA Indonesia (CC BY-SA 4.0) [4], GEO-TREES [5], Labrière *et al.* (CC0) [6], and the SAFE tree census [8]. Satellite data were read from the Microsoft Planetary Computer [32] and Google Earth Engine. The pipeline (Python; pandas, LightGBM, rasterio, pystac) is in the project repository, including the ledger, the runner, the right-grid builder, and the evaluation scripts. Every reported number is regenerated by the modules `right_grid`, `right_grid_seq`, `right_grid_uncertainty`, `extract_left`, `evaluate`, `evaluate_seq`, `improve`, `improve_seq`, `window_sensitivity`, `cloud_climatology`, `cloud_analysis`, `robust`, `project_uncertainty`, and `systems_experiments` of the `estimation` package. Wood density is from the GWDD copy shipped with BIOMASS v2.1.11 [11].

References

[1] J. Chave *et al.*, "Improved allometric models to estimate the aboveground biomass of tropical trees," *Global Change Biol.*, vol. 20, no. 10, pp. 3177–3190, 2014, doi: 10.1111/gcb.12629.

[2] Verra, *VM0047 Afforestation, Reforestation, and Revegetation*, ver. 1.1, May 2025. [Online]. Available: https://verra.org/wp-content/uploads/2025/05/VM0047-v1.1_Final.pdf

[3] G. Rosoman, S. S. Sheun, C. Opal, P. Anderson, and R. Trapshah, Eds., *The HCS Approach Toolkit*, ver. 2.0, Module 4. Singapore: HCS Approach Steering Group, 2017.

[4] High Carbon Stock Approach, "Forest field plot data for indicative HCS mapping, Indonesia," Fair Forward/GIZ and JKPP, 2023, CC BY-SA 4.0. [Online]. Available: https://huggingface.co/datasets/fair-forward/hcsa_indonesia

[5] GEO-TREES, "Biomass reference plot data," IIASA, accessed Oct. 2026. [Online]. Available: https://geo-trees.nodes.iiasa.ac.at/page/data

[6] N. Labrière *et al.*, "In situ reference datasets from the TropiSAR and AfriSAR campaigns in support of upcoming spaceborne biomass missions," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 10, pp. 3617–3627, 2018, doi: 10.1109/JSTARS.2018.2851606.

[7] R. M. Ewers *et al.*, "A large-scale forest fragmentation experiment: The Stability of Altered Forest Ecosystems Project," *Philos. Trans. R. Soc. B*, vol. 366, pp. 3292–3302, 2011, doi: 10.1098/rstb.2011.0049.

[8] M. Svátek *et al.*, "Tree census data from the SAFE Project 2011–2020," Zenodo, 2025, doi: 10.5281/zenodo.14882506.

[9] A. E. Zanne *et al.*, "Data from: Towards a worldwide wood economics spectrum," Dryad, 2009, doi: 10.5061/dryad.234.

[10] J. Chave, D. Coomes, S. Jansen, S. L. Lewis, N. G. Swenson, and A. E. Zanne, "Towards a worldwide wood economics spectrum," *Ecol. Lett.*, vol. 12, no. 4, pp. 351–366, 2009, doi: 10.1111/j.1461-0248.2009.01285.x.

[11] M. Réjou-Méchain, A. Tanguy, C. Piponiot, J. Chave, and B. Hérault, "BIOMASS: An R package for estimating above-ground biomass and its uncertainty in tropical forests," *Methods Ecol. Evol.*, vol. 8, no. 9, pp. 1163–1167, 2017.

[12] T. A. P. West *et al.*, "Action needed to make carbon offsets from forest conservation work for climate change mitigation," *Science*, vol. 381, no. 6660, pp. 873–877, 2023.

[13] R. Dubayah *et al.*, "The Global Ecosystem Dynamics Investigation: High-resolution laser ranging of the Earth's forests and topography," *Sci. Remote Sens.*, vol. 1, Art. no. 100002, 2020.

[14] L. Duncanson *et al.*, "Aboveground biomass density models for NASA's Global Ecosystem Dynamics Investigation (GEDI) lidar mission," *Remote Sens. Environ.*, vol. 270, Art. no. 112845, 2022.

[15] M. Santoro *et al.*, "The global forest above-ground biomass pool for 2010 estimated from high-resolution satellite observations," *Earth Syst. Sci. Data*, vol. 13, pp. 3927–3950, 2021.

[16] S. A. Spawn, C. C. Sullivan, T. J. Lark, and H. K. Gibbs, "Harmonized global maps of above and belowground biomass carbon density in the year 2010," *Sci. Data*, vol. 7, Art. no. 112, 2020.

[17] Chloris Geospatial, "Global aboveground biomass stock and change, 2003–2019," distributed by the Microsoft Planetary Computer. [Online]. Available: https://planetarycomputer.microsoft.com/dataset/chloris-biomass

[18] E. T. A. Mitchard *et al.*, "Using satellite radar backscatter to predict above-ground woody biomass: A consistent relationship across four different African landscapes," *Geophys. Res. Lett.*, vol. 36, Art. no. L23401, 2009.

[19] S. Mermoz, M. Réjou-Méchain, L. Villard, T. Le Toan, V. Rossi, and S. Gourlet-Fleury, "Decrease of L-band SAR backscatter with biomass of dense forests," *Remote Sens. Environ.*, vol. 159, pp. 307–317, 2015.

[20] G. P. Asner and J. Mascaro, "Mapping tropical forest carbon: Calibrating plot estimates to a simple LiDAR metric," *Remote Sens. Environ.*, vol. 140, pp. 614–624, 2014.

[21] G. Ke *et al.*, "LightGBM: A highly efficient gradient boosting decision tree," in *Adv. Neural Inf. Process. Syst. 30*, 2017, pp. 3146–3154.

[22] N. Duan, "Smearing estimate: A nonparametric retransformation method," *J. Amer. Statist. Assoc.*, vol. 78, no. 383, pp. 605–610, 1983.

[23] D. R. Roberts *et al.*, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, pp. 913–929, 2017.

[24] P. Ploton *et al.*, "Spatial validation reveals poor predictive performance of large-scale ecological mapping models," *Nat. Commun.*, vol. 11, Art. no. 4540, 2020.

[25] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York, NY, USA: Springer, 2005.

[26] L. I.-K. Lin, "A concordance correlation coefficient to evaluate reproducibility," *Biometrics*, vol. 45, no. 1, pp. 255–268, 1989.

[27] S. Haber and W. S. Stornetta, "How to time-stamp a digital document," *J. Cryptology*, vol. 3, no. 2, pp. 99–111, 1991.

[28] M. Blum, "Coin flipping by telephone: A protocol for solving impossible problems," *ACM SIGACT News*, vol. 15, no. 1, pp. 23–27, 1983.

[29] J. Dean and S. Ghemawat, "MapReduce: Simplified data processing on large clusters," *Commun. ACM*, vol. 51, no. 1, pp. 107–113, 2008.

[30] D. Karger, E. Lehman, T. Leighton, R. Panigrahy, M. Levine, and D. Lewin, "Consistent hashing and random trees," in *Proc. 29th ACM Symp. Theory Comput.*, 1997, pp. 654–663.

[31] R. L. Graham, "Bounds on multiprocessing timing anomalies," *SIAM J. Appl. Math.*, vol. 17, no. 2, pp. 416–429, 1969.

[32] Microsoft Open Source, M. McFarland, R. Emanuele, D. Morris, and T. Augspurger, "microsoft/PlanetaryComputer: October 2022," Zenodo, 2022, doi: 10.5281/zenodo.7261897.

[33] IPCC, *2006 IPCC Guidelines for National Greenhouse Gas Inventories, Vol. 4: Agriculture, Forestry and Other Land Use*, Ch. 2 and 4, H. S. Eggleston *et al.*, Eds. Hayama, Japan: IGES, 2006.

[34] IPCC, *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Vol. 4*, Ch. 4 Forest Land, Table 4.4 (Updated). Geneva: IPCC, 2019.

[35] IPCC, *2006 IPCC Guidelines for National Greenhouse Gas Inventories, Vol. 1*, Ch. 3 Uncertainties. Hayama, Japan: IGES, 2006.

[36] Badan Standardisasi Nasional, *SNI 7724:2011 Pengukuran dan penghitungan cadangan karbon – Pengukuran lapangan untuk penaksiran cadangan karbon hutan (ground based forest carbon accounting)*. Jakarta: BSN, 2011.

[37] Ministry of Environment and Forestry, Republic of Indonesia, *National Forest Reference Level for Deforestation, Forest Degradation and Enhancement of Forest Carbon Stock*, submission to the UNFCCC, 2022. [Online]. Available: https://redd.unfccc.int/files/2nd_frl_indonesia_final_submit.pdf

[38] UNFCCC CDM Executive Board, *A/R Methodological Tool 14: Estimation of Carbon Stocks and Change in Carbon Stocks of Trees and Shrubs in A/R CDM Project Activities*, ver. 04.2.

[39] UNFCCC CDM Executive Board, *A/R Methodological Tool 03: Calculation of the Number of Sample Plots for Measurements within A/R CDM Project Activities*, ver. 2.1.0, EB 58, Annex 15.

[40] Gold Standard, *Afforestation/Reforestation GHG Emissions Reduction & Sequestration Methodology*, ver. 2.1, May 2024.

[41] Verra, *VMD0017 Estimation of Uncertainty for REDD+ Project Activities*, ver. 2.2, Sep. 2020.

[42] J. Chave, R. Condit, S. Aguilar, A. Hernandez, S. Lao, and R. Perez, "Error propagation and scaling for tropical forest biomass estimates," *Phil. Trans. R. Soc. B*, vol. 359, pp. 409–420, 2004, doi: 10.1098/rstb.2003.1425.

[43] T. M. Basuki, P. E. van Laake, A. K. Skidmore, and Y. A. Hussin, "Allometric equations for estimating the above-ground biomass in tropical lowland Dipterocarp forests," *Forest Ecol. Manag.*, vol. 257, pp. 1684–1694, 2009, doi: 10.1016/j.foreco.2009.01.027.

[44] R. C. Goodman *et al.*, "Amazon palm biomass and allometry," *Forest Ecol. Manag.*, vol. 310, pp. 994–1004, 2013, doi: 10.1016/j.foreco.2013.09.045.

[45] K. Lewis *et al.*, "An assessment of oil palm plantation aboveground biomass stocks on tropical peat using destructive and non-destructive methods," *Sci. Rep.*, vol. 10, art. 2230, 2020, doi: 10.1038/s41598-020-58982-9.

[46] S. C. Thomas and A. R. Martin, "Carbon content of tree tissues: A synthesis," *Forests*, vol. 3, no. 2, pp. 332–352, 2012, doi: 10.3390/f3020332.

[47] M. Réjou-Méchain *et al.*, "Local spatial structure of forest biomass and its consequences for remote sensing of carbon stocks," *Biogeosciences*, vol. 11, pp. 6827–6840, 2014, doi: 10.5194/bg-11-6827-2014.

[48] A. M. Wilson and W. Jetz, "Remotely sensed high-resolution global cloud dynamics for predicting ecosystem and biodiversity distributions," *PLoS Biol.*, vol. 14, no. 3, e1002415, 2016, doi: 10.1371/journal.pbio.1002415.

[49] M. Sudmanns, D. Tiede, H. Augustin, and S. Lang, "Assessing global Sentinel-2 coverage dynamics and data availability for operational Earth observation (EO) applications using the EO-Compass," *Int. J. Digit. Earth*, vol. 13, no. 7, pp. 768–784, 2020, doi: 10.1080/17538947.2019.1572799.

[50] J. Li and D. P. Roy, "A global analysis of Sentinel-2A, Sentinel-2B and Landsat-8 data revisit intervals and implications for terrestrial monitoring," *Remote Sens.*, vol. 9, no. 9, art. 902, 2017, doi: 10.3390/rs9090902.

[51] Z. Zhu and C. E. Woodcock, "Object-based cloud and cloud shadow detection in Landsat imagery," *Remote Sens. Environ.*, vol. 118, pp. 83–94, 2012, doi: 10.1016/j.rse.2011.10.028.

[52] S. Skakun *et al.*, "Cloud Mask Intercomparison eXercise (CMIX): An evaluation of cloud masking algorithms for Landsat 8 and Sentinel-2," *Remote Sens. Environ.*, vol. 274, art. 112990, 2022, doi: 10.1016/j.rse.2022.112990.

[53] V. J. Pasquarella, C. F. Brown, W. Czerwinski, and W. J. Rucklidge, "Comprehensive quality assessment of optical satellite imagery using weakly supervised video learning," in *Proc. IEEE/CVF CVPR Workshops*, 2023, pp. 2125–2135, doi: 10.1109/CVPRW59228.2023.00206.

[54] J. C. White *et al.*, "Pixel-based image compositing for large-area dense time series applications and science," *Can. J. Remote Sens.*, vol. 40, no. 3, pp. 192–212, 2014, doi: 10.1080/07038992.2014.945827.

[55] M. Claverie *et al.*, "The Harmonized Landsat and Sentinel-2 surface reflectance data set," *Remote Sens. Environ.*, vol. 219, pp. 145–161, 2018, doi: 10.1016/j.rse.2018.09.002.

[56] J. Reiche *et al.*, "Combining satellite data for better tropical forest monitoring," *Nature Clim. Change*, vol. 6, pp. 120–122, 2016, doi: 10.1038/nclimate2919.

[57] A. Meraner, P. Ebel, X. X. Zhu, and M. Schmitt, "Cloud removal in Sentinel-2 imagery using a deep residual neural network and SAR-optical data fusion," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 333–346, 2020, doi: 10.1016/j.isprsjprs.2020.05.013.

[58] N. Neverova, C. Wolf, G. Taylor, and F. Nebout, "ModDrop: Adaptive multi-modal gesture recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, 2016, doi: 10.1109/TPAMI.2015.2461544.

[59] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, "Distribution-free predictive inference for regression," *J. Amer. Statist. Assoc.*, vol. 113, no. 523, pp. 1094–1111, 2018, doi: 10.1080/01621459.2017.1307116.

[60] Y. Romano, E. Patterson, and E. Candès, "Conformalized quantile regression," in *Adv. Neural Inf. Process. Syst. 32*, 2019, pp. 3543–3553.

[61] H. Papadopoulos, A. Gammerman, and V. Vovk, "Normalized nonconformity measures for regression conformal prediction," in *Proc. IASTED Int. Conf. Artificial Intelligence and Applications*, 2008, pp. 64–69.

[62] H. Boström and U. Johansson, "Mondrian conformal regressors," in *Proc. COPA*, PMLR vol. 128, 2020, pp. 114–133.

[63] R. J. Tibshirani, R. F. Barber, E. Candès, and A. Ramdas, "Conformal prediction under covariate shift," in *Adv. Neural Inf. Process. Syst. 32*, 2019, pp. 2530–2540.

[64] H. Mao, R. Martin, and B. J. Reich, "Valid model-free spatial prediction," *J. Amer. Statist. Assoc.*, vol. 119, no. 546, pp. 904–914, 2024, doi: 10.1080/01621459.2022.2147531.

[65] L. Duncanson, M. Disney, J. Armston, J. Nickeson, D. Minor, and F. Camacho, Eds., *Aboveground Woody Biomass Product Validation Good Practices Protocol*, ver. 1.0, CEOS WGCV Land Product Validation Subgroup, 2021, doi: 10.5067/doc/ceoswgcv/lpv/agb.001.

[66] L. Duncanson *et al.*, "The importance of consistent global forest aboveground biomass product validation," *Surv. Geophys.*, vol. 40, no. 4, pp. 979–999, 2019, doi: 10.1007/s10712-019-09538-8.

[67] A. Araza *et al.*, "A comprehensive framework for assessing the accuracy and uncertainty of global above-ground biomass maps," *Remote Sens. Environ.*, vol. 272, art. 112917, 2022, doi: 10.1016/j.rse.2022.112917.

[68] A. Berninger, S. Lohberger, M. Stängel, and F. Siegert, "SAR-based estimation of above-ground biomass and its changes in tropical forests of Kalimantan using L- and C-band," *Remote Sens.*, vol. 10, no. 6, art. 831, 2018, doi: 10.3390/rs10060831.

[69] T. Jucker *et al.*, "Estimating aboveground carbon density and its uncertainty in Borneo's structurally complex tropical forests using airborne laser scanning," *Biogeosciences*, vol. 15, pp. 3811–3830, 2018, doi: 10.5194/bg-15-3811-2018.

[70] W. G. Cochran, *Sampling Techniques*, 3rd ed. New York: Wiley, 1977.

[71] C.-E. Särndal, B. Swensson, and J. Wretman, *Model Assisted Survey Sampling*. New York: Springer, 1992.

[72] M. Zaffran, A. Dieuleveut, J. Josse, and Y. Romano, "Conformal prediction with missing values," in *Proc. 40th Int. Conf. Machine Learning (ICML)*, PMLR vol. 202, 2023, pp. 40578–40604.

[73] G. Ståhl *et al.*, "Use of models in large-area forest surveys: Comparing model-assisted, model-based and hybrid estimation," *Forest Ecosyst.*, vol. 3, art. 5, 2016, doi: 10.1186/s40663-016-0064-9.

[74] E. Rutishauser *et al.*, "Generic allometric models including height best estimate forest biomass and carbon stocks in Indonesia," *Forest Ecol. Manag.*, vol. 307, pp. 219–225, 2013, doi: 10.1016/j.foreco.2013.07.013.

[75] K. Mokany, R. J. Raison, and A. S. Prokushkin, "Critical analysis of root:shoot ratios in terrestrial biomes," *Global Change Biol.*, vol. 12, pp. 84–96, 2006, doi: 10.1111/j.1365-2486.2005.001043.x.

[76] A. Blum and M. Hardt, "The Ladder: A reliable leaderboard for machine learning competitions," in *Proc. 32nd Int. Conf. Machine Learning (ICML)*, PMLR vol. 37, 2015, pp. 1006–1014.

[77] A. I. Flores-Anderson *et al.*, "Spatial and temporal availability of cloud-free optical observations in the tropics to monitor deforestation," *Sci. Data*, vol. 10, art. 550, 2023, doi: 10.1038/s41597-023-02439-x.

[78] M. Kuronen, J. Räty, P. Packalen, and M. Myllymäki, "Uncertainty quantification for forest attribute maps with conformal prediction and k-nearest neighbor method," *Remote Sens. Environ.*, vol. 325, art. 114758, 2025, doi: 10.1016/j.rse.2025.114758.

[79] G. Singh, G. Moncrieff, Z. Venter, K. Cawse-Nicholson, J. Slingsby, and T. B. Robinson, "Uncertainty quantification for probabilistic machine learning in Earth observation using conformal prediction," arXiv:2401.06421, 2024.

[80] E. W. Mauya *et al.*, "Effects of field plot size on prediction accuracy of aboveground biomass in airborne laser scanning-assisted inventories in tropical rain forests of Tanzania," *Carbon Balance Manag.*, vol. 10, art. 10, 2015, doi: 10.1186/s13021-015-0021-x.

[81] J. Breidenbach, R. E. McRoberts, and R. Astrup, "Empirical coverage of model-based variance estimators for remote sensing assisted estimation of stand-level timber volume," *Remote Sens. Environ.*, vol. 173, pp. 274–281, 2016, doi: 10.1016/j.rse.2015.07.026.

[82] A. M. J.-C. Wadoux and G. B. M. Heuvelink, "Scientists yet to consider spatial correlation in assessing uncertainty of spatial averages and totals," *Int. J. Appl. Earth Obs. Geoinf.*, vol. 139, art. 104472, 2025, doi: 10.1016/j.jag.2025.104472.

[83] H. Meyer and E. Pebesma, "Predicting into unknown space? Estimating the area of applicability of spatial prediction models," *Methods Ecol. Evol.*, vol. 12, pp. 1620–1633, 2021, doi: 10.1111/2041-210X.13650.

[84] G. Grassi *et al.*, "Applying the conservativeness principle to REDD to deal with the uncertainties of the estimates," *Environ. Res. Lett.*, vol. 3, art. 035005, 2008, doi: 10.1088/1748-9326/3/3/035005.

[85] I. Shishlov and V. Bellassen, "Review of the experience with monitoring uncertainty requirements in the Clean Development Mechanism," *Climate Policy*, vol. 16, pp. 703–731, 2016, doi: 10.1080/14693062.2015.1046414.

[86] G. Badgley *et al.*, "Systematic over-crediting in California's forest carbon offsets program," *Global Change Biol.*, vol. 28, pp. 1433–1445, 2022, doi: 10.1111/gcb.15943.

[87] D. Requena Suarez *et al.*, "Estimating aboveground net biomass change for tropical and subtropical forests: Refinement of IPCC default rates using forest plot data," *Global Change Biol.*, vol. 25, pp. 3609–3624, 2019, doi: 10.1111/gcb.14767.